Born-Valid Dissipative Self-Interacting Dark Matter and the Compact Lens JVAS B1938+666

[Author names withheld for review]

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Abstract

We test whether the compact strong-lens perturber in JVAS B1938+666 provides evidence for dissipative self-interacting dark matter (dSIDM) when the interaction is tied to a controlled particle-physics model. We combine Born-limit velocity-dependent transfer cross sections with the energy-differential emission kernels for long-range identical-fermion χ–V and χ–ϕ scattering. Their emitted-energy moment is thermally averaged over the relative-speed distribution and implemented as a cooling source in a spherical gravothermal fluid solver. We impose an explicit Born-validity condition, evolve isolated Navarro–Frenk–White halos, and use the two projected masses measured at 20 and 90 pc in B1938+666. Since the microscopic velocity scale breaks the constant-cross-section rescaling symmetry, every selected profile is rerun directly with its inferred physical halo parameters.

Our screening scan covers two emission channels. We vary η_B from 0.03 to 0.3, the reference cross section from 0.1 to 10 $\mathrm{cm^2\,g^{-1}}$, and the threshold velocity from 20 to 500 $\mathrm{km\,s^{-1}}$. A uniform 192-shell time scan of the final six candidates gives a minimum two-mass statistic $\chi^2 = 0.01772$ at the unevolved boundary. The directly resimulated halos have a maximum one-dimensional velocity dispersion of only $5.37\,\mathrm{km\,s^{-1}}$. Even for $v_\star = 20\,\mathrm{km\,s^{-1}}$, the exact thermal emission moment implies a minimum local cooling time of 1.9×10^{26}–2.3×10^{26} Gyr. At a common evolved source snapshot, dissipative and cooling-disabled controls have identical projected masses and χ^2 at the saved precision. Thus B1938+666 can be matched by the rescaled initial or effectively elastic halo, but the tested Born-valid models do not generate an observable dissipative signature. This is a null test of dissipation in the explored microphysical regime, not an exclusion of dSIDM in general.

1 Introduction

The standard collisionless ΛCDM model reproduces the large-scale matter distribution, but galaxy-scale observations motivate tests of its assumptions about dark-matter microphysics. The central-density, diversity, and too-big-to-fail discussions are not each a unique failure of ΛCDM, but together they motivate models in which dark matter can exchange momentum and heat (Bullock & Boylan-Kolchin, 2017). Self-interacting dark matter (SIDM) is the minimal realization of this idea: elastic scattering redistributes energy inside a halo without introducing a new luminous component. Early cosmological calculations established the resulting core formation and changes to halo structure (Spergel & Steinhardt, 2000; Davé et al., 2001; Colín et al., 2002), while later simulations showed how the effect depends on halo mass, concentration, and the required cross section (Elbert et al., 2015; Garrison-Kimmel et al., 2013; Tulin & Yu, 2018).

The astrophysical interpretation is necessarily velocity dependent. A constant cross section per unit mass is a useful benchmark, but light-mediator models naturally produce transfer cross sections that vary strongly with relative velocity (Tulin et al., 2013; Kaplinghat et al., 2016). Resonant and long-range regimes can therefore soften dwarf-galaxy tensions while avoiding the much tighter limits from clusters. The Bullet Cluster remains a direct high-velocity test (Randall et al., 2008), and cluster strong lensing and group kinematics provide complementary constraints on the same velocity-dependent interaction (Andrade et al., 2022; Kahlhoefer et al., 2014; Sagunski et al., 2021). Baryons and the angular structure of scattering can also alter the mapping from a microscopic cross section to a halo profile (Robertson et al., 2017; Shen et al., 2021). These issues are now sufficiently developed that SIDM is best treated as a constrained particle-astrophysics framework rather than as a single constant-σ/m model (Adhikari et al., 2025).

Elastic scattering alone already leads to gravothermal evolution. Heat flows outward from a hot center, a core can form transiently, and the subsequent loss of pressure support can drive contraction or core collapse (Balberg et al., 2002; Koda & Shapiro, 2011; Nishikawa et al., 2020; Outmezguine et al., 2023). If a scattering event also radiates energy into a dark-sector state, the cooling channel competes with conductive transport and can accelerate this sequence. Dissipative dark-sector constructions have been studied in the context of dark disks, bound states, and compact-object formation (Fan et al., 2013; Wise & Zhang, 2014; Essig et al., 2019; Xiao et al., 2021). A phenomenological cooling coefficient is useful for isolating this dynamical mechanism, but it does not guarantee that a particle model can realize the assumed cooling rate at the velocities present in the halo.

Strong gravitational lensing supplies an unusually clean way to test this question. A small perturber is detected through its gravitational potential, so the inference does not require a luminous counterpart, and gravitational imaging can constrain projected mass on tens-of-parsecs scales. This is the role of the compact perturber in JVAS B1938+666: a roughly million-solar-mass object at the lens redshift $z = 0.881$ was identified through its effect on the lensed surface brightness (Powell et al., 2025; Vegetti et al., 2026). The inferred profile is summarized particularly efficiently by two projected masses, inside 20 and 90 pc, which constrain both the central normalization and the profile shape. It is therefore a useful case study for gravothermal models, not an assumption that the perturber has already been proven to be dSIDM.

Using velocity-independent transfer and dissipation parameters, Schmidt et al. (2026) showed that both elastic and weakly dissipative SIDM halos can reproduce these two masses near the late stages of gravothermal evolution. Their phenomenological result establishes a conditional possibility: dissipation can reduce the evolution time or, equivalently, the cross section required to obtain a compact halo. The remaining question is whether the same conclusion survives when the cooling and scattering rates are tied to a controlled microscopic model.

A light mediator produces a velocity-dependent transfer cross section, while emission of a massive boson introduces a threshold in relative velocity. These scales break the rescaling symmetry of a constant-cross-section halo and can suppress dissipation in low-velocity systems. In addition, perturbative cross sections are predictive only inside their domain of validity. The relevant question is therefore not whether a prescribed constant cooling parameter can affect a halo, but whether a Born-valid microscopic model populates the radiative region after the B1938 halo has been placed in physical units.

We address this question with three ingredients. First, we implement the Born-limit elastic transfer cross section and the long-range identical-fermion emission kernels derived by Lankester-Broche & Pradler (2026). Second, we perform an explicit Born-valid parameter scan rather than assigning an independent cooling strength. Third, we use the constant-model rescaling only to propose physical initial conditions and then rerun every candidate directly, preserving its microscopic velocity scale. The result is a self-consistent test of whether B1938 probes dissipation in this class of models.

This paper is organized as follows. Section 2 describes the scattering models and cooling moment. Section 3 summarizes the fluid solver, B1938 statistic, and scan strategy. Section 4 presents the Born-valid parameter map, direct 192-shell time scan, physical threshold audit, and elastic controls. We discuss the physical interpretation and limitations in Section 5 and conclude in Section 6.

2 Microphysical framework

2.1 Velocity-dependent scattering and Born validity

For identical particles of mass m_χ and reduced mass $\mu = m_\chi/2$, we use the Born-limit transfer cross section

$$\sigma_T(v_{\rm rel}) = \frac{a^4}{8\pi\mu^2 v_{\rm rel}^4}\left[\ln\left(1+\frac{4\mu^2 v_{\rm rel}^2}{m_{\rm med}^2}\right) - \frac{4\mu^2 v_{\rm rel}^2/m_{\rm med}^2}{1+4\mu^2 v_{\rm rel}^2/m_{\rm med}^2}\right], \tag{1}$$

where $m_{\rm med}$ is the mediator mass and a denotes the effective coupling entering the scattering amplitude. The cross section approaches a constant at small velocity and falls approximately as $v_{\rm rel}^{-4}$, up to a logarithm, at large velocity.

The perturbative expansion is monitored with

$$\eta_B = \frac{\alpha_D \mu}{m_{\rm med}}. \tag{2}$$

We classify $\eta_B < 1$ as Born-valid and use $\eta_B = 0.1$ for the final controlled candidates. The emission threshold is parameterized by

$$v_\star = \sqrt{\frac{2m_{\rm med}}{\mu}}. \tag{3}$$

For a chosen $(\eta_B, v_\star)$, the dark-matter and mediator masses are varied together to obtain a target value of σ_T/m_χ at 100 km s^{-1}.

We study two emission channels. Model M1 contains massive-vector emission in long-range χ–V scattering, while M2 contains massive-scalar emission in long-range χ–ϕ scattering. The elastic parameters can be chosen to give the same reference cross section and threshold, but the two channels retain different energy-differential emission kernels. This construction separates differences caused by radiation from differences in the elastic normalization.

2.2 Energy-weighted cooling moment

Let ω be the radiated energy in a collision. For each relative speed we compute

$$Q(v_{\rm rel}) = \int_{m_{\rm em}}^{\mu v_{\rm rel}^2/2} d\omega\, \omega \frac{d\sigma}{d\omega}, \tag{4}$$

using the long-range identical-fermion kernels of Lankester-Broche & Pradler (2026). The corresponding cooling combination is

$$\left(\frac{\sigma}{m_\chi}\right)_{\rm cool}(v_{\rm rel}) = \frac{2Q(v_{\rm rel})}{m_\chi^2 v_{\rm rel}^2}. \tag{5}$$

The volumetric energy-loss rate for one identical population is

$$C_{\rm vol} = \frac{\rho^2}{4}\left\langle \left(\frac{\sigma}{m_\chi}\right)_{\rm cool}(v_{\rm rel}) v_{\rm rel}^3 \right\rangle_{\rm rel}, \tag{6}$$

where the factor 1/4 accounts for identical-particle pairs in the adopted normalization. The relative speed follows a Maxwell distribution whose component variance is twice the one-particle one-dimensional variance. This average includes the high-velocity tail and is therefore more informative than a sharp comparison between a single characteristic velocity and $v_\star$.

For constant σ_T/m_χ and a constant fractional dissipation parameter $r_{\rm diss}$, Eq. (6) reduces to

$$C_{\rm vol} = \frac{8}{\sqrt{\pi}}\frac{\sigma_T}{m_\chi}(r_{\rm diss}-1)\rho^2\nu^3, \tag{7}$$

where ν is the one-dimensional velocity dispersion. This is the constant-parameter limit used by Schmidt et al. (2026). In our microphysical model, the energy moment fixes the dissipative loss rather than an independent constant parameter.

3 Numerical method

3.1 Gravothermal evolution

We evolve spherical, isolated Navarro–Frenk–White (NFW) halos (Navarro et al., 1997) with the GravothermalSIDM fluid equations (Balberg et al., 2002; Koda & Shapiro, 2011; Nishikawa et al., 2020; Outmezguine et al., 2023; Gad-Nasr et al., 2024). The initial profile has scale radius $r_s = 3.6$ kpc and scale density $\rho_s = 7.09 \times 10^{-3}\, M_\odot\, {\rm pc}^{-3}$. The solver follows Lagrangian mass shells in hydrostatic equilibrium and interpolates between long- and short-mean-free-path heat conduction. At each shell, the elastic transport coefficient is thermally averaged from Eq. (1) and the specific cooling rate is $C_{\rm vol}/\rho$ from Eq. (6). The timestep limits both conductive and radiative changes in the specific internal energy.

The calculation is performed with 96 shells for parameter screening and 192 shells for the final candidates. A 48-shell grid was also tested, but it can shift the preferred snapshot and produce spuriously small mass residuals; it is therefore used only for software-level checks. The final timestep tolerance is $\epsilon_t = 0.005$.

3.2 B1938 projected masses

We adopt the B1938 projected masses

$$M_{20}^{\rm obs} = (4.25 \pm 0.21) \times 10^5\, M_\odot, \tag{8}$$
$$M_{90}^{\rm obs} = (1.167 \pm 0.039) \times 10^6\, M_\odot, \tag{9}$$

at projected radii of 20 and 90 pc (Vegetti et al., 2026). The corresponding observed ratio is 0.364 ± 0.022. Projected masses are calculated by integrating the exact cylinder-intersection volume of each piecewise-constant fluid shell.

For every physical direct run we use the two-mass statistic

$$\chi^2_{2M} = \left(\frac{M_{20} - M_{20}^{\rm obs}}{\delta M_{20}} \right)^2 + \left(\frac{M_{90} - M_{90}^{\rm obs}}{\delta M_{90}} \right)^2 . \tag{10}$$

This statistic is a compact diagnostic based on two measurements; it is not a replacement for the full lens-imaging likelihood.

3.3 Direct physical resimulation

For constant cross sections, the transformation

$$r \to \lambda r, \qquad M \to \mu M, \qquad t \to \left(\frac{\lambda^3}{\mu} \right)^{1/2} t, \qquad \frac{\sigma}{m} \to \frac{\lambda^2}{\mu} \frac{\sigma}{m} \tag{11}$$

maps one gravothermal solution into another (Schmidt et al., 2026). A massive emission threshold is not invariant under this transformation because the velocity changes as $(\mu/\lambda)^{1/2}$. We therefore use Eq. (11) only to infer a candidate physical r_s, ρ_s, and target time. A new 192-shell halo is then evolved with those physical parameters and the unchanged microscopic model. The final masses in Eq. (10) are measured from this direct run, without a post-evolution mass rescaling.

The source-snapshot grid for the final run is

$$t_{\rm source}/{\rm Gyr} = 0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.10. \tag{12}$$

The explicit zero-time point is included to determine whether the best fit is an evolved state or a boundary initial condition. The physical target time of a direct run is reported separately from $t_{\rm source}$.

3.4 Physical activity audit and controls

For each direct halo we compare the maximum physical one-dimensional dispersion with the emission threshold and evaluate the shortest local cooling time,

$$t_{\rm cool}^{\rm min} = t_{\rm scale} \min_i \left(\frac{u_i}{C_{{\rm cool},i}} \right), \tag{13}$$

where u_i and $C_{{\rm cool},i}$ are in code units. The exact thermal kernel remains in Eq. (13), so a possible Maxwell tail above threshold is retained. Finally, the six final candidates are rerun at $t_{\rm source} = 0.05$ Gyr with the cooling term disabled but identical velocity-dependent elastic transport.

The numerical implementation is covered by 28 regression tests. They check the relative-speed distribution, the constant-parameter limit of Eq. (6), the massless analytic emission rate, threshold closure of the massive kernels, Born-parameter construction, exact shell projection, and direct-snapshot selection.

Table 1: Numerical scans used in the analysis. The final direct scan contains two emission models at each of three threshold velocities and seven source times.

Stage	shells	η_B	$\sigma_T/m_\chi(100)$ $[\mathrm{cm^2\,g^{-1}}]$	$v_\star$ $[\mathrm{km\,s^{-1}}]$
Born screening	96	$0.03, 0.1, 0.3$	$0.1, 1, 10$	100
Final time scan	192	0.1	0.1	$20, 100, 500$
Elastic control	192	0.1	0.1	$20, 100, 500$

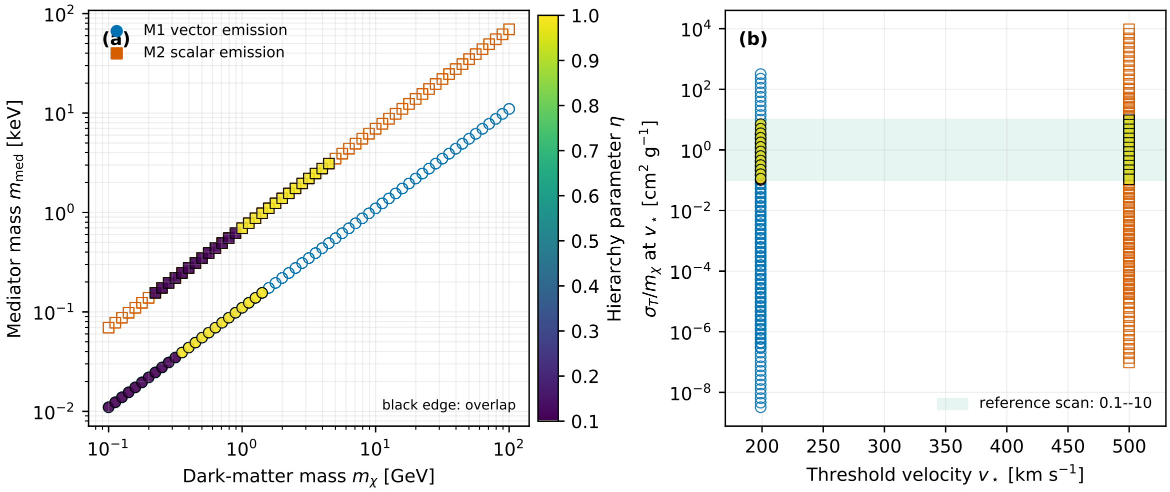

Figure 1: Born-valid mass-hierarchy screening. (a) Dark-matter and mediator masses in the hierarchy scan; colors encode η_B, and black-edged points satisfy the astrophysical overlap criterion. (b) Transfer cross section per mass at the imposed threshold velocity. The shaded band marks the 0.1–10 $\mathrm{cm^2\,g^{-1}}$ reference range used for the controlled scan.

4 Results

4.1 Born-valid parameter space

The Born condition substantially restricts a direct interpretation of large phenomenological dSIDM cross sections. If only the coupling is varied at the nominal M1 and M2 mass hierarchies, the inferred expansion parameters are much larger than unity. At the limiting value $\eta_B = 1$, the largest transfer cross sections at 100 $\mathrm{km\,s^{-1}}$ are approximately 3.23×10^{-4} and $9.72 \times 10^{-3}\,\mathrm{cm^2\,g^{-1}}$ for M1 and M2, respectively. Thus a fixed normalization of order $50\,\mathrm{cm^2\,g^{-1}}$ cannot be combined with these nominal masses in the Born regime.

A Born-valid astrophysical window opens when the mass hierarchy is varied at fixed threshold. For $\eta_B = 0.1$ and $0.1 \le \sigma_T/m_\chi(100\,\mathrm{km\,s^{-1}}) \le 10\,\mathrm{cm^2\,g^{-1}}$, the corresponding dark-matter masses are approximately 0.10–0.32 GeV for M1 and 0.22–0.89 GeV for M2 at their nominal thresholds. This motivates a controlled scan in which η_B, the reference cross section, and $v_\star$ are treated as explicit coordinates.

Figure 1 visualizes the mass-hierarchy scan behind this selection. The filled points satisfy the astrophysical overlap criterion, whereas open points are Born-valid but outside that reference window. The direct calculations therefore sample a controlled subset of the broader microphysical construction rather than treating every Born-valid point as an equally plausible B1938 candidate.

Table 2: Direct candidates in the $v_\star = 100\,\mathrm{km\,s^{-1}}$ screening branch. The source time is selected from the saved fluid snapshots and is not the physical direct-evolution time.

$\sigma_T/m_\chi(100)$ [$\mathrm{cm^2\,g^{-1}}$]	t_{source} [Gyr]	χ^2_{2M}	M_{20}/M_{90}
0.1	0.0500	0.0256	0.3677
1	0.500	0.0909	0.3578
10	0.300	0.5816	0.3478

Table 3: Direct two-mass statistic in the final 192-shell scan. M1 and M2 are identical at the reported precision.

t_{source} [Gyr]	$\chi^2_{2M}(v_\star = 20)$	$\chi^2_{2M}(v_\star = 100)$	$\chi^2_{2M}(v_\star = 500)$
0	0.01772	0.01772	0.01772
0.01	0.01803	0.01823	0.01903
0.05	0.01929	0.02034	0.02476
0.10	0.02093	0.02315	0.03309

4.2 The 100 km/s threshold branch

The 96-shell $v_\star = 100\,\mathrm{km\,s^{-1}}$ scan contains 18 completed fluid evolutions and 18 direct physical candidate runs. The result is independent of emission channel and nearly independent of η_B at fixed reference cross section. Table 2 lists the representative direct result for each cross section.

The improved fit at small cross section is not caused by radiation. After the physical rescaling, all 18 direct halos have $v_{\max} = 5.23$–$5.38\,\mathrm{km\,s^{-1}}$, so the characteristic dispersion is less than 5.4 percent of the imposed threshold. The thermally averaged massive-emission kernel is numerically zero in these states. The variation in Table 2 is therefore an elastic-transport trend.

4.3 Uniform 192-shell time scan

We select the lowest-cross-section controlled point with $\eta_B = 0.1$. Its reference cross section is $0.1\,\mathrm{cm^2\,g^{-1}}$ at $100\,\mathrm{km\,s^{-1}}$. We then perform the uniform 192-shell time scan for both models and $v_\star = 20, 100, 500\,\mathrm{km\,s^{-1}}$. All 42 direct runs complete. The M1 and M2 values agree exactly at the saved precision, so Table 3 reports one value per threshold.

Figure 2(a) shows the full time grid. The statistic increases monotonically away from the initial state for all three threshold velocities. The global minimum is

$$\chi^2_{2M} = 0.0177167, \qquad \frac{M_{20}}{M_{90}} = 0.3670805, \tag{14}$$

at $t_{\mathrm{source}} = 0$. It corresponds to zero physical evolution time and zero conduction steps. Equation (14) therefore demonstrates that the rescaled initial NFW halo can match the two projected masses; it is not evidence for gravothermal or dissipative evolution.

The nonzero source times map to much shorter physical times. At the largest source time in Eq. (12), the direct target time is only 1.10×10^{-3} Gyr, or approximately 1.1 Myr. The increase of χ^2_{2M} with t_{source} is controlled by elastic evolution and is strongest for the largest threshold branch because fixing the threshold and reference cross section also fixes a different velocity dependence of the elastic transport.

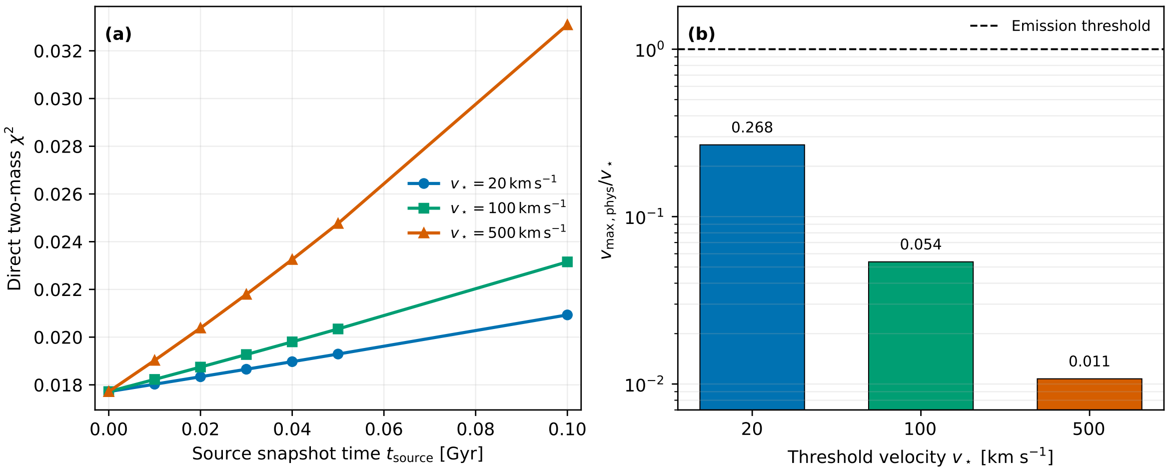

Figure 2: Final physical B1938 audit. (a) Direct two-mass statistic along the uniform 192-shell source-time grid. M1 and M2 overlap at the plotted precision. (b) Maximum physical one-dimensional velocity dispersion divided by the emission threshold. The dashed line marks unity; all candidates lie well below it.

Table 4: Physical activity audit of the final candidates. An infinite cooling time denotes a numerically vanishing thermally averaged emission kernel.

$v_\star$ [km s^{-1}]	$v_{\rm max}$ [km s^{-1}]	$v_{\rm max}/v_\star$	$t_{\rm cool}^{\rm min}$ M1 [Gyr]	$t_{\rm cool}^{\rm min}$ M2 [Gyr]
20	5.367	0.268	1.91×10^{26}	2.29×10^{26}
100	5.368	0.0537	∞	∞
500	5.371	0.0107	∞	∞

4.4 Physical threshold and cooling audit

The physical velocity hierarchy is summarized in Table 4. The maximum direct dispersion is approximately $5.37\,{\rm km\,s^{-1}}$ for every candidate. The smallest threshold is still 3.7 times larger than this dispersion; the 100 and 500 km s^{-1} thresholds are larger by factors of 19 and 93.

For $v_\star = 20\,{\rm km\,s^{-1}}$, the relative-speed Maxwell tail is included by construction, but the resulting cooling is dynamically irrelevant. The largest cooling diagnostic over the direct scan is 3.7×10^{-31} in code units for M1 and 2.9×10^{-31} for M2. For the two higher thresholds it is zero at numerical precision.

The cooling-disabled controls provide a direct test of the elastic limit. At $t_{\rm source} = 0.05$ Gyr, all six controls use the same physical initial conditions, elastic transport, resolution, and timestep tolerance as their dissipative counterparts. For every pair we find

$$\Delta\chi^2_{2M} = 0, \qquad \Delta(M_{20}/M_{90}) = 0 \tag{15}$$

at the saved precision, with identical conduction-step counts. The equality in Eq. (15) verifies that the final candidate set is in the effective elastic limit.

5 Discussion

5.1 From phenomenological cooling to microscopic thresholds

Velocity-independent dSIDM calculations establish an important conditional result: if the local cooling strength is appreciable, dissipation can change the gravothermal trajectory and accelerate the production of compact halos (Schmidt et al., 2026). Our calculation asks whether a specific perturbative emission model realizes that condition in the physical B1938 halo. It does not in the parameter region tested here.

This difference is not a disagreement about the fluid response to cooling. Equation (6) reproduces the constant-model source term in Eq. (7). The difference arises before the nonlinear halo response: the physical velocity distribution scarcely populates the massive-emission region, and the exact microscopic cooling moment is negligible. The phenomenological and microscopic calculations therefore probe different dynamical regimes.

The Born audit is central to this interpretation. A large independent value of $r_{\mathrm{diss}} - 1$ can be assigned in a model-agnostic fluid calculation, but the scattering and emission rates are linked once masses and couplings are specified. At fixed nominal mass hierarchy, the large cross sections often used to produce rapid gravothermal evolution lie outside the Born expansion. Moving to a Born-valid hierarchy can recover astrophysically relevant elastic cross sections, but does not guarantee that the radiative threshold is active in the compact physical halo.

5.2 What B1938 does and does not imply

The very small value in Eq. (14) should not be interpreted as a detection of dSIDM. It is obtained at the unevolved boundary, where all emission models are identical and no interaction history has accumulated. The evolved dissipative runs are also indistinguishable from the elastic controls. Within this analysis, B1938 therefore supplies no positive evidence for radiative self-interactions.

The result also does not exclude dSIDM in general. The observational input is compressed to two projected masses, the initial halo is restricted to an isolated NFW family, and the microphysical scan covers two long-range identical-fermion channels. A model with a lower threshold, a different emission channel, a nonperturbative scattering treatment, or a halo modified by tides and assembly history could yield a different cooling hierarchy. A full statistical claim would additionally require the lens-imaging likelihood and priors on the halo mass, concentration, and formation time.

The present null result is nevertheless physically informative. It identifies a necessary condition for using B1938 as a dSIDM probe: after the candidate is placed in physical units, its relative-speed distribution must produce a cooling time comparable to the available evolution time. Source-frame threshold activation or a successful constant-model rescaling is insufficient.

5.3 Numerical scope

The final results use the spherical gravothermal approximation. This is appropriate for the controlled comparison of transport and local cooling in an isolated halo, but it does not include triaxiality, mergers, tidal stripping, or stochastic particle dynamics. Independent N-body validation would be required in a parameter region where cooling measurably changes the fluid trajectory. For the present candidates, however, the decisive hierarchy is kinematic and the exact cooling time exceeds any astrophysical time by more than 20 orders of magnitude. An N-body calculation cannot turn this inactive microscopic source into an observable cooling signal.

The shell-resolution study also shows that coarse grids can alter the selected snapshot. The 48-shell results are not used for quantitative statements; screening is performed at 96 shells and the final time scan and elastic controls at 192 shells. A broader parameter search should retain this separation between screening and final resolution.

6 Conclusions

We have performed a self-consistent gravothermal test of Born-valid, velocity-dependent dSIDM in the compact strong-lens perturber JVAS B1938+666. The main conclusions are:

1. The constant-parameter cooling law is recovered as a limit of the emitted-energy moment, while massive vector and scalar emission introduce a fixed microscopic velocity scale.

2. At nominal mass hierarchies, large phenomenological dSIDM cross sections are incompatible with a controlled Born expansion. Varying the hierarchy opens a Born-valid elastic window at lower dark-matter masses.

3. The final 192-shell scan is minimized at the unevolved boundary, $\chi^2_{2M} = 0.01772$. This is a successful initial-profile fit, not a dissipative-evolution signal.

4. The directly resimulated physical halos have $v_{\rm max} \simeq 5.37\,{\rm km\,s^{-1}}$, well below all three tested thresholds. The smallest finite cooling time is of order 10^{26} Gyr.

5. Cooling-disabled and dissipative direct runs give identical B1938 masses at fixed physical parameters. The tested models are therefore in the effective elastic limit.

The B1938 mass measurements do not favor dissipative self-interactions in the explored Born-valid parameter region, but they do not exclude other dSIDM regimes. Future work should first identify a microscopic branch that is both theoretically controlled and radiatively active in the physical halo; only then would broader high-resolution or N-body calculations be warranted.

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